

LED Equations:

$$\text{LED1} = A + B + C + D$$

$$\text{LED2} = AB + BC + CD + AC + AD + BD$$

$$\text{LED3} = ABC + ABD + ACD + BCD$$

$$\text{LED4} = ABCD$$

* Warning LED $\rightarrow$ Same as LED 3

$$ABC + ABD + ACD + BCD$$

7 Segment Display Equations:

Segment (a): [Segment a and d have same equations]

$$AB'(C+D) + BC'(A+D) + C(BD' + A'D) + A'B'C'D'$$

Segment (b): 1 [Always Active]

Segment c:

$$AB(c+d) + cD(A+B) + B'D'(A'+c') + A'c'(B'+D')$$

Segment d: [Same as Segment a]

$$AB'(c+d) + BC'(A+D) + C(BD' + A'D) + A'B'c'D'$$

Segment e:

$$A'B'(c \odot D) + (c \oplus D)(A \oplus B) + ABC'D'$$

Segment f : $ABCD + A'B'C'D'$

Segment g : [Same as LED-02]

$$AB + BC + CD + AC + AD + BD$$